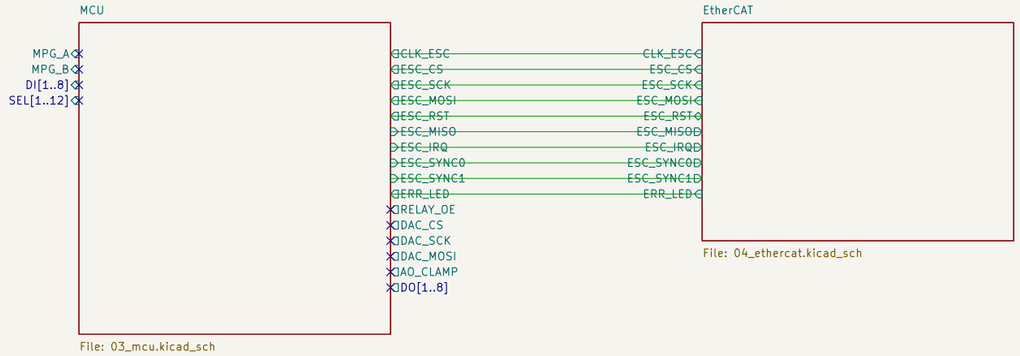
LCEC-IO — EtherCAT slave I/O board

Root sheet, milestone M4: Power (2), MCU (3) and EtherCAT (4). Sheets 5 to 8 (digital inputs, MPG, relay outputs, analog outputs) are added in M5; the full root sheet follows in M6.





No-connect flags on the MCU sheet pins are placeholders: they are replaced by wires to sheets 5 to 8 in M5.

Root sheet: Power, MCU and EtherCAT sheets

Yantra Innovation — LCEC-IO

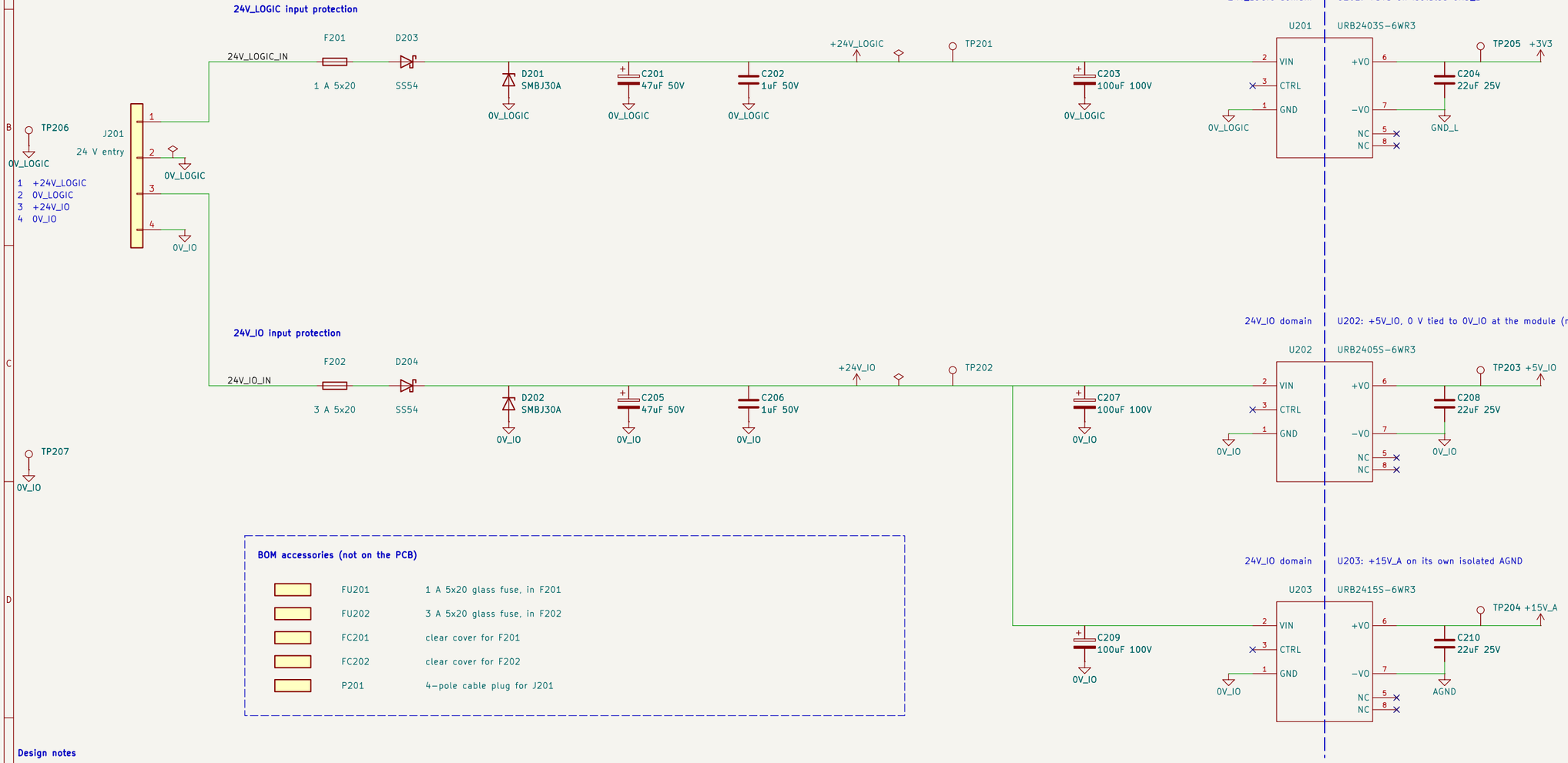
Sheet: /
File: lcec-io.kicad_sch

Title: LCEC-IO — EtherCAT slave I/O board

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/4

2 POWER – 24 V entries, protection, three isolated DC–DC modules

Input 24 VDC +/–20 % = 19.2 … 28.8 V. Per input: fuse > SS54 reverse–polarity diode > SMBJ30A TVS > 47 uF + 1 uF, then a 100 uF/100 V input capacitor at each URB module (URB datasheet Cin).



BOM accessories (not on the PCB)

	FU201	1 A 5x20 glass fuse, in F201
	FU202	3 A 5x20 glass fuse, in F202
	FC201	clear cover for F201
	FC202	clear cover for F202
	P201	4–pole cable plug for J201

Design notes

Loads: +3V3 about 0.4 A of the module's 1.35 A (30 %). 24V_LOGIC input about 0.3 A at full module load: 1 A fuse.

24V_IO: 5 V module 7.3 W + 15 V module 7 W at full load = 0.75 A, plus relay coils 0.21 A and sensors through the PTCs 1 A: 3 A fuse.

Input inrush I2t into the input capacitors stays below 1.1 A2s; the fuses are rated 4.26 A2s (1 A) and 12.5 A2s (3 A).

URB input 9–36 V (40 V continuous, 50 V for 1 s). SMBJ30A: 30 V stand–off above the 28.8 V maximum input, clamps 48.4 V at 12.4 A.

CTRL left open = module on. NC pins are not connected inside the module.

Two 24 V entries, protection, three isolated DC–DC modules

Yantra Innovation – LCEC–IO

Sheet: /Power/

File: 02_power.kicad_sch

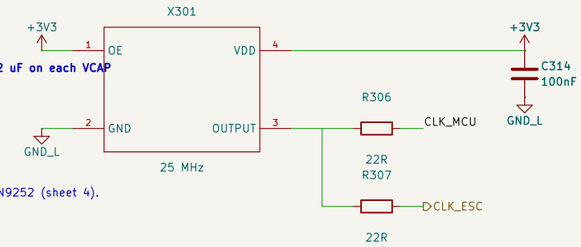
Title: LCEC–IO – 2 Power

Size: A3	Date: 2026–11–06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/4

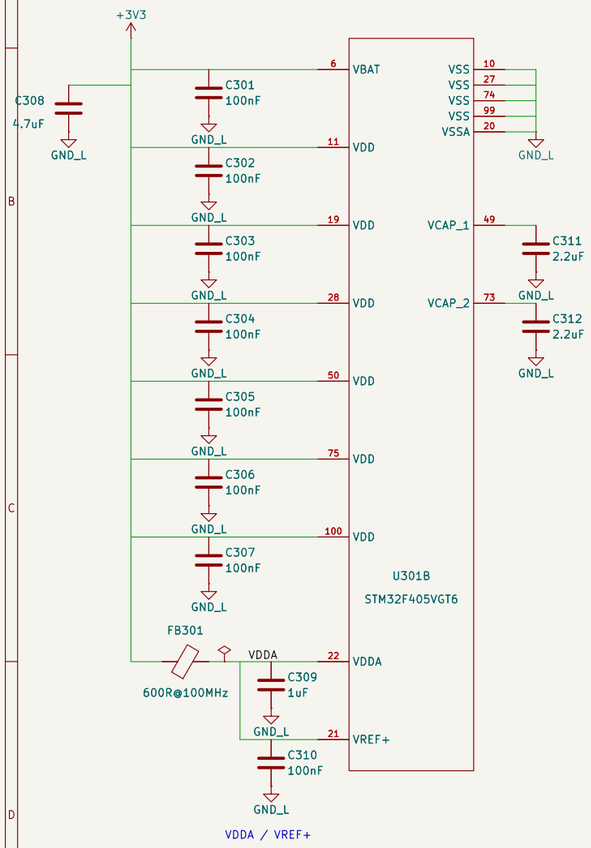
3 MCU — STM32F405VGT6, shared 25 MHz clock, SWD, UART2, heartbeat LED

51 signals used (pin map in docs/plan.md), PH1 is a test pad, the 30 pins of ports D and E are free for future use (no-connect flags). Firmware: HSE bypass, PLLM 25 / PLLN 336 / PLLP 2 = 168 MHz, IWDG enabled in option bytes.

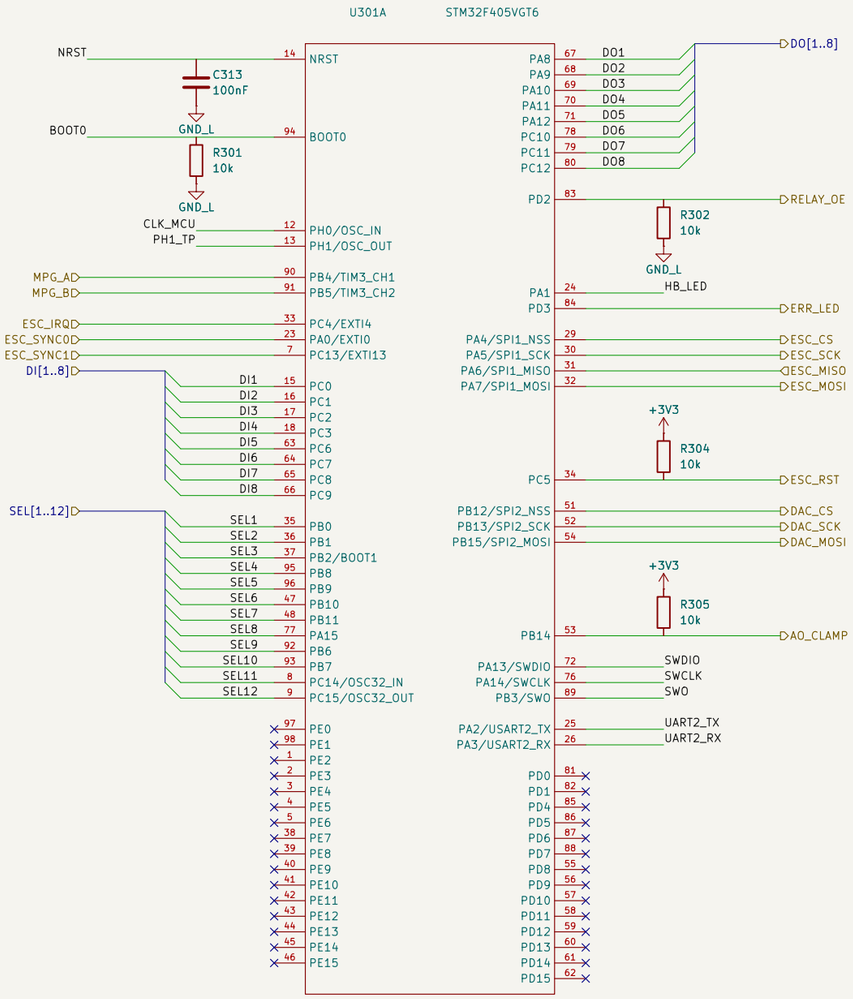
U301 power: 100 nF beside each VDD pin and VBAT, 4.7 uF bulk, VDDA through a ferrite, 2.2 uF on each VCAP



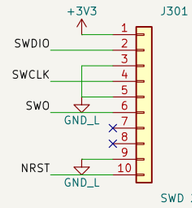
OE tied high = oscillator running. 22 R series resistors to PH0 (HSE bypass) and to the LAN9252 (sheet 4).



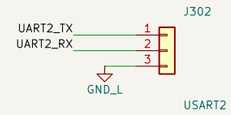
VDDA / VREF+



SWD (Cortex 10-pin): 1 VTref, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7/8 not used, 10 NRST



USART2: 1 TX (PA2), 2 RX (PA3), 3 GND



Test points: SPI1 (LAN9252), SPI2 (DAC), relay OE, analog clamp, MPG A/B and the spare pin PH1





STM32F405VGT6 (LQFP-100), shared clock, SWD, UART, heartbeat LED

Yantra Innovation — LCEC-IO

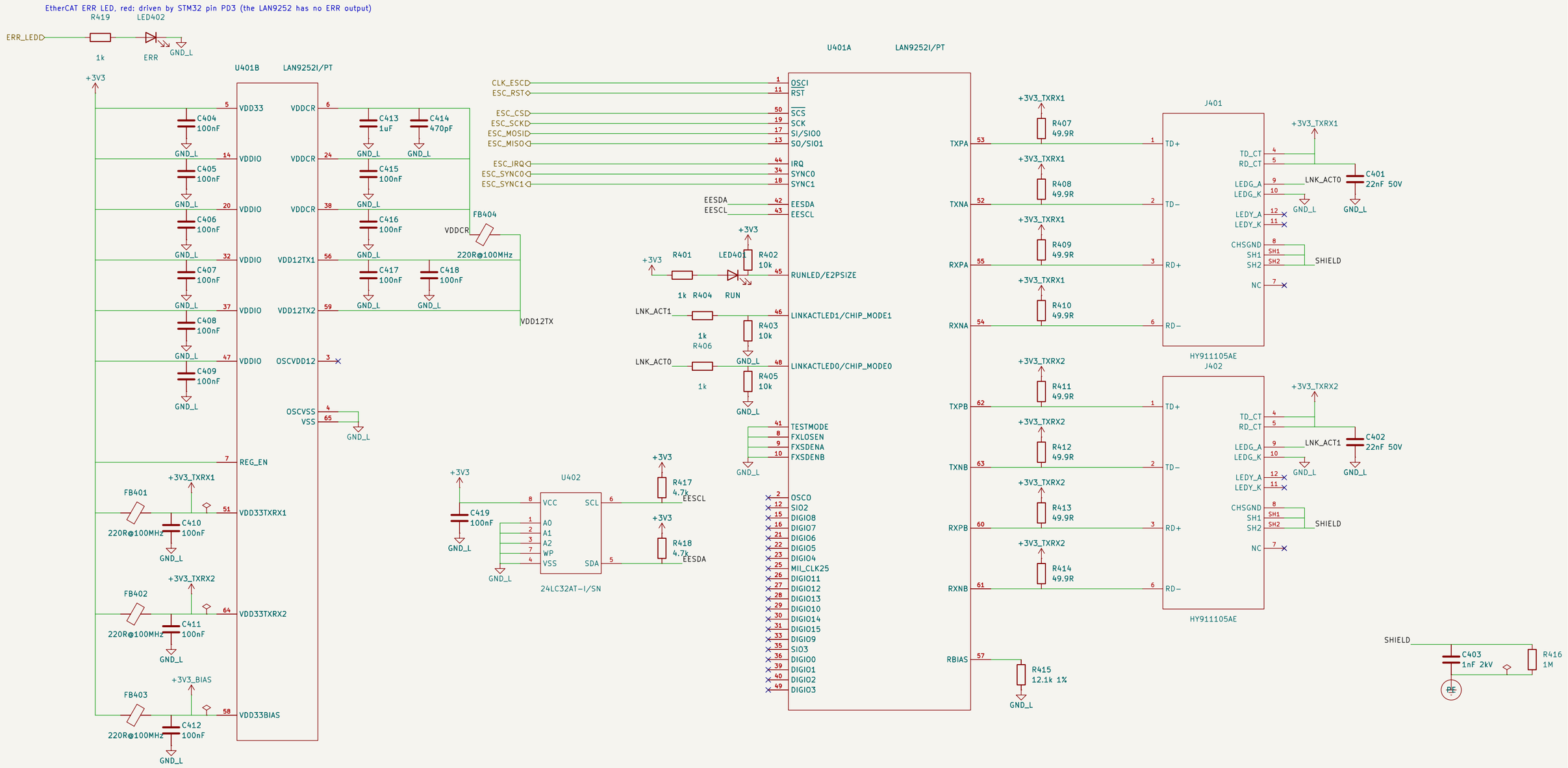
Sheet: /MCU/
File: 03_mcu.kicad_sch

Title: LCEC-IO — 3 MCU

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/4

4 ETHERCAT — LAN9252 (SPI), ESI EEPROM, two RJ45 jacks with magnetics, RUN and ERR LEDs

Chip mode 00 = 2 ports (PHY A = port 0 = J401, PHY B = port 1 = J402). The PDI is chosen by the ESI EEPROM content (SPI slave). Rows and values follow the LAN9252 datasheet DS00001909A (sections 3, 4, 7, 12.10 and 14).



Design notes

RUNLED / E2PSIZE: 10 k pull-up = EEPROM size strap high (32 Kbit and up, 2-byte addressing). The pin is open drain, so the RUN LED is +3V3 > 1 k > LED > pin.

LINKACTLED0 / CHIP_MODE0 and LINKACTLED1 / CHIP_MODE1: 10 k pull-downs = chip mode 00 (2 ports) and active-high LED outputs; 1 k to the green LED in each jack.

TESTMODE to ground (required). FXLOSEN, FXSDENA and FXSDENB to ground = copper twisted pair. REG_EN to 3.3 V = internal 1.2 V regulators. OSCO open (single-ended oscillator on OSCI).

Pins 12, 15 to 16, 21 to 23, 25 to 31, 33, 35, 36, 39, 40 and 49 (host bus, MII, digital I/O) are unused in SPI mode and left open.

As in the EVB-LAN9252-SPI reference: VDD33TXRX1, VDD33TXRX2 and VDD33BIAS each come from +3V3 through a 220 R ferrite bead with 100 nF; VDD12TX1/2 come from VDDCR through a ferrite bead with 2 x 100 nF.

Each twisted-pair line has 49.9 R to its port's filtered 3.3 V; the magnetics centre taps (TXCT and RXCT together) are on the same net with 22 nF. RBIAS 12.1 k 1 % to ground. VDDCR: 1 uF + 470 pF at pin 6, 100 nF at pins 24 and 38.

The ESI EEPROM must hold PDI = SPI before the first run. WP is low so the master can write it (the EVB ties WP high).

LAN9252 in SPI mode, 24LC32A ESI EEPROM, two RJ45 jacks, RUN and ERR LEDs

Yantra Innovation — LCEC-IO

Sheet: /EtherCAT/

File: 04_ethercat.kicad_sch

Title: LCEC-IO — 4 EtherCAT

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/4